

NovaStream Model Validation

Purpose and context

A proof-of-concept case study comparing classification approaches in a premium-upgrade scenario. This exploration informed the approach to Healthcare AI Analytics; it is not a standalone application or a healthcare model validation.

Approach

- Check data quality and compare baseline groups.
- Compare Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.
- Review feature importance and produce upgrade-probability scores.

Documented evidence

- Model comparison workflow
- Feature-importance reporting
- Decision-support scoring

Outcome and interpretation

The saved evaluation compares four classifiers using multiple metrics. Reviewing recall alongside accuracy exposes tradeoffs that a single headline score would hide.

Scope and limitations

Premium-upgrade proof of concept that informed the Healthcare AI Analytics approach. These results are not healthcare or clinical validation. Customer-level workbooks and scoring records remain private.

Evidence status

Prepared from curated project records. This document summarizes existing work; it is not a fresh test execution or independent certification. The separate evidence report identifies supporting records and any outstanding validation.